



UniUyo ID

Digital student ID cards with QR verification

The university already has a register of students with names and registration numbers. UniUyo ID turns each record into a digital ID card with a signed QR code. A lecturer scans the card and immediately sees whether the person is a UniUyo student, and whether they belong to the class being taught.

48

students in the register

7

courses, 64 registrations

36

past scans

Contents

1. Overview
2. How the core logic works
3. Architecture and data model
4. Screen walkthrough
5. Mobile view
6. Running locally
7. Testing
8. Deployment
9. Five minute presentation script

Demo logins: lecturer@uniuyo.edu.ng / lecturer123, student@uniuyo.edu.ng / student123, uduak.etuk@student.uniuyo.edu.ng / student123. Screens captured with the clock frozen at 2026-10-05.

1. Overview

Who	What they do
Student	Logs in and sees their ID card (front and back) with photo, registration number, department, level, expiry and QR code, plus their registered courses. Can print the card.
Lecturer	Chooses one of their courses and scans a card with the camera, uploads a photo of the QR, or types a registration number. Sees a green, amber or red answer with the student's details. Sees the class list and who was verified today.
Anyone	Scanning the QR with a normal phone camera opens a public page that confirms the card is genuine and current, without showing course details.

The three questions a scan answers

1. Is this card genuine? (the QR signature is checked)
2. Is this a UniUyo student in good standing? (register, card version, suspension, expiry)
3. Does this student belong to this class? (course registration, then department)

Design

The interface follows the visual style of ChatGPT: white canvas, a light grey sidebar with history, Inter type, a rounded composer box with tool buttons, and results shown like messages in a thread. It is a normal web app, not a chatbot.

2. How the core logic works

Signed QR codes

```
QR text: https://<site>/verify/UU1.<base64url reg no>.<card version>.<signature>
signature = first 16 bytes of HMAC-SHA256(server secret, "UU1.<reg>.<version>")
```

Changing one character of the registration number or version breaks the signature, so a student cannot make a card for someone else. The signature is compared in constant time. When a card is reissued, `card_version` goes up and QR codes on the old card stop working.

Verdict rules (`src/lib/verify.ts`)

Check, in order	Result	Lecturer sees
Signature fails	invalid	Not a UniUyo ID
Reg. number not in register	unknown	Student not found
Card version is old	revoked	Old card
Status suspended	suspended	Student suspended
Expired or graduated	expired	ID card expired
Registered for the course	in_class	Belongs to CSC 311
Same department, not registered	same_department	Not registered for CSC 311
Different department	other_department	UniUyo student, not in this class

Every check is saved in the scans table with the lecturer, course and result, so the class list can show who was verified today. Lecturers can only scan for, and see rosters of, courses they teach.

Reading the QR

The camera scanner uses `getUserMedia` and the `jsQR` library to read frames in the browser. The same decoder reads uploaded photos. Typed registration numbers are normalised (upper case, single slashes) and looked up directly; the result says "Looked up by registration number" because no card was presented.

Dates

Rules take "today" as an argument. `UNIID_TODAY=YYYY-MM-DD` freezes the clock for demos and tests. Cards expiring within 30 days show a warning.

3. Architecture and data model

Browser

Server-rendered pages. Client components: scanner (camera, jsQR), sidebar toggle, login demo buttons.

Next.js 16

Server Components, a Server Action for each scan, QR SVGs rendered on the server, proxy.ts route guard, /api/health.

PostgreSQL

Drizzle ORM, SQL migrations from drizzle-kit.

Table	Columns	Rules
students	id, reg_no, first_name, last_name, other_name, gender, faculty, department, level, entry_year, status, id_issued_on, id_expires_on, card_version, blood_group	reg_no unique; the existing register
users	id, name, email, password_hash, role (student, lecturer), department, student_id	email unique; student_id links a login to a register entry
courses	id, code, title, department, level, units, venue, lecturer_id	code unique
enrollments	id, student_id, course_id	unique (student_id, course_id)
scans	id, lecturer_id, course_id, student_id, result, scanned_at	result is one of the eight verdicts

Route	Who	Purpose
/login	All	Log in, demo accounts
/card	Student	ID card, status and courses
/scan	Lecturer	Scanner and results thread
/courses/[id]	Lecturer	Class list with today's checks
/verify, /verify/[token]	Public	Card check for anyone
/api/health	Railway	Health check with database ping

4. Screen walkthrough

Log in



Welcome back

Log in to UniUyo ID to show your card or verify students in class.

1

Email address

lecturer@uniuyo.edu.ng

2

Password

.....

2

Continue

3

 **Lecturer: Dr. Aniefiok Ekong**
lecturer@uniuyo.edu.ng / lecturer123

 **Student in CSC 311: Ubong Akpan**
student@uniuyo.edu.ng / student123

 **Law student: Uduak Etuk**
uduak.etuk@student.uniuyo.edu.ng / student123



4

One scan, three answers.
Is this card genuine, is this a UniUyo student, and are they registered for this class?

Log-in page.

- 1 Email and password, prefilled with the lecturer demo account.
- 2 Continue checks the bcrypt hash and sets a signed JWT session cookie (HTTP-only).
- 3 One click fills in the lecturer, a CSC 311 student, or a Law student for the 'not in this class' case.
- 4 Real lecture theatre photo (Creative Commons).

Student ID card

UniUyo ID

My ID card

Registered courses

About

Tools

Public card check

My ID card

Print card

Hi Ubong, here is your student ID

Lecturers scan the QR code to confirm you are a UniUyo student and check your course registration.

Valid until 23 Apr 2027.

UNIVERSITY OF UYO

STUDENT IDENTITY CARD

AKPAN Ubong

Daniel

REG. NUMBER

J/23/CSC/045

DEPARTMENT

Computer Science

LEVEL

300 Level

Faculty of Science

Expires 23 Apr 2027

IF FOUND, PLEASE RETURN

REGISTRAR, UNIVERSITY OF UYO, AKWA IBOM

Card details

Issued: 15 Nov 2023 · Card version 1

Entry year: 2023 · Gender: Male · Blood group: O+

This card remains the property of the University of Uyo. Scan the QR code to confirm it is genuine and current.

Registrar

Registered courses this semester

Code	Title	Units	Lecturer	Venue
CSC 305	Database Systems	3	Dr. Aniefiok Ekong	ICT Centre, Lab 1
CSC 311	Operating Systems	3	Dr. Aniefiok Ekong	Faculty of Science LT 2

Ubong Akpan

Student

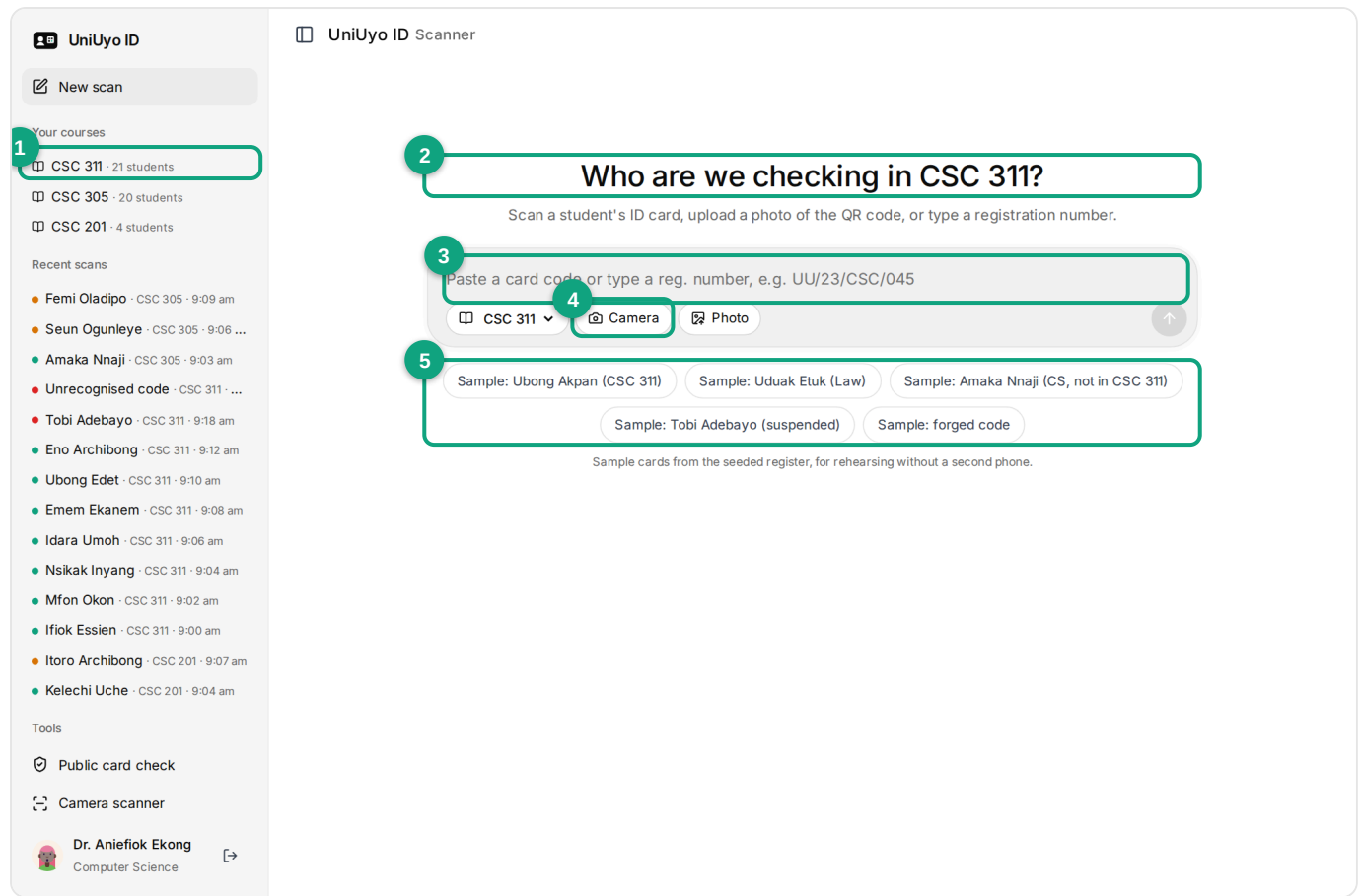
The demo student's card.

- 1 Card status from `cardState()`: valid, expiring within 30 days, expired or suspended.
- 2 ID photo: a locally generated portrait (DiceBear), never a real person's face.
- 3 Name, registration number, department and level come from the student register.
- 4 QR code with a signed verification link. It cannot be forged or edited without the server key.
- 5 Courses this student is registered for, so they know which classes the scan will pass.

UniUyo ID documentation

Page 6 of 17

Lecturer scanner



Scanner before any scan.

1. Sidebar like a chat history: the lecturer's courses and recent scans, coloured by result.
2. The selected course is part of every check.
3. Paste the card code or type a registration number such as UU/23/CSC/045.
4. Camera scanner (jsQR reads frames from the phone or laptop camera) and Photo upload.
5. Sample cards from the seeded register for rehearsing with one device.

Scan results

The screenshot displays the UniUyo ID Scanner interface. On the left is a sidebar with navigation options: 'New scan', 'Your courses' (listing CSC 311, CSC 305, and CSC 201), 'Recent scans' (a list of past scans with student names and times), 'Tools' (Public card check, Camera scanner), and a user profile for Dr. Aniefiok Ekong. The main area is titled 'UniUyo ID Scanner' and contains a text input field for a card code or registration number, with a dropdown menu set to 'CSC 311'. Below this are buttons for 'Camera' and 'Photo'. A green box labeled '1' highlights the input field containing 'UU/23/CSC/045'. A green box labeled '2' highlights a green status bar that says 'Belongs to CSC 311'. Below this is a student profile for 'AKPAN Ubong Daniel', registered for CSC 311, with details: Reg. no UU/23/CSC/045, Level 300, Dept Computer Science, and Expires 23 Apr 2027. A sample student 'Uduak Etuk (Law)' is also shown. A green box labeled '3' highlights an amber status bar that says 'UniUyo student, not in this class'. Below this is a student profile for 'ETUK Uduak', from the Department of Law, with details: Reg. no UU/23/LAW/112, Level 300, Dept Law, and Expires 16 Jul 2027. A green box labeled '4' highlights a green status bar at the bottom that says 'Checked from QR signature · UniUyo student'.

UniUyo ID

New scan

Your courses

- CSC 311 · 21 students
- CSC 305 · 20 students
- CSC 201 · 4 students

Recent scans

- Ubong Akpan · CSC 311 · 1:52 am
- Uduak Etuk · CSC 311 · 1:52 am
- Femi Oladipo · CSC 305 · 9:09 am
- Seun Ogunleye · CSC 305 · 9:06 ...
- Amaka Nnaji · CSC 305 · 9:03 am
- Unrecognised code · CSC 311 · ...
- Tobi Adebayo · CSC 311 · 9:18 am
- Eno Archibong · CSC 311 · 9:12 am
- Ubong Edet · CSC 311 · 9:10 am
- Emem Ekanem · CSC 311 · 9:08 am
- Idara Umoh · CSC 311 · 9:06 am
- Nsikak Inyang · CSC 311 · 9:04 am
- Mfon Okon · CSC 311 · 9:02 am
- Ifiok Essien · CSC 311 · 9:00 am

Tools

- Public card check
- Camera scanner

Dr. Aniefiok Ekong
Computer Science

UniUyo ID Scanner

Paste a card code or type a reg. number, e.g. UU/23/CSC/045

CSC 311 Camera Photo

1 UU/23/CSC/045

2 Belongs to CSC 311

AKPAN Ubong Daniel
Registered for CSC 311 this semester.
Reg. no UU/23/CSC/045 Level 300
Dept Computer Science Expires 23 Apr 2027

CSC 311 Operating Systems · Looked up by registration number · UniUyo student

Sample: Uduak Etuk (Law)

3 UniUyo student, not in this class

ETUK Uduak
Department of Law. Not on the CSC 311 class list.
Reg. no UU/23/LAW/112 Level 300
Dept Law Expires 16 Jul 2027

4 CSC 311 Operating Systems · Checked from QR signature · UniUyo student

A Law student, then a CSC 311 student, scanned for CSC 311. Newest on top.

- 1 What was scanned, shown like a sent message.
- 2 Green: genuine card and registered for CSC 311.
- 3 Amber: a real UniUyo student, but from the Law department and not on this class list.
- 4 How it was checked (QR signature or reg. number) and whether the person is a confirmed UniUyo student.

Class list

UniUyo ID

New scan

Your courses

CSC 311 · 21 students

CSC 305 · 20 students

CSC 201 · 4 students

Recent scans

Ubong Akpan · CSC 311 · 1:52 am

Uduak Etuk · CSC 311 · 1:52 am

Femi Oladipo · CSC 305 · 9:09 am

Seun Ogunleye · CSC 305 · 9:06 ...

Amaka Nnaji · CSC 305 · 9:03 am

Unrecognised code · CSC 311 · ...

Tobi Adebayo · CSC 311 · 9:18 am

Eno Archibong · CSC 311 · 9:12 am

Ubong Edet · CSC 311 · 9:10 am

Emem Ekanem · CSC 311 · 9:08 am

Idara Umoh · CSC 311 · 9:06 am

Nsikak Inyang · CSC 311 · 9:04 am

Mfon Okon · CSC 311 · 9:02 am

Ifiok Essien · CSC 311 · 9:00 am

Tools

Public card check

Camera scanner

Dr. Aniefiok Ekong
Computer Science

CSC 311 class list

4 Scan for this class

CSC 311: Operating Systems

Computer Science · 300 Level · 3 units · Faculty of Science LT 2

1 21 registered students

1 verified in class today

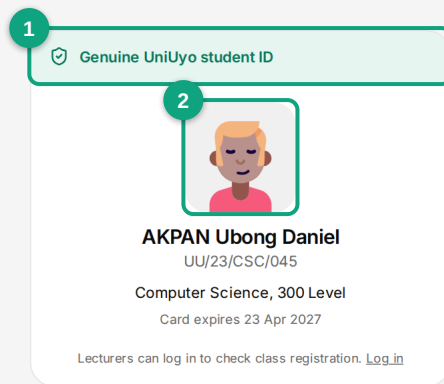
3 flags (outsiders today, card problems)

Student	Reg. number	Card	Last verified
 ABUBAKAR Halima Computer Science	UU/23/CSC/078	3 expiring	Not yet
 ADEBAYO Tobi Computer Science	UU/23/CSC/081	2 suspended	Not yet
 ADEYEMI Tunde Computer Science	UU/23/CSC/056	Valid	Not yet
 AKPAN Ubong Daniel Computer Science	UU/23/CSC/045	Valid	Today
 AKPANUDO Samuel Electrical/Electronic Engineering	UU/23/EEE/310	Valid	Not yet
 ARCHIBONG Eno Computer Science	UU/23/CSC/040	Valid	3 Oct, 9:12 am
 BASSEY Ekemini Computer Science	UU/23/CSC/012	Valid	26 Sept, 9:00 am
 EDET Ubong Computer Science	UU/23/CSC/038	Valid	3 Oct, 9:10 am
 EKANEM Emem Computer Science	UU/23/CSC/034	Valid	3 Oct, 9:08 am
 ESSIEN Ifiok			

CSC 311 class list.

- 1 Class size, students verified today, and flags (outsiders scanned today plus card problems).
- 2 A suspended student on the class list.
- 3 A card that expires within 30 days.
- 4 Jump to the scanner with this course selected.

Public card check

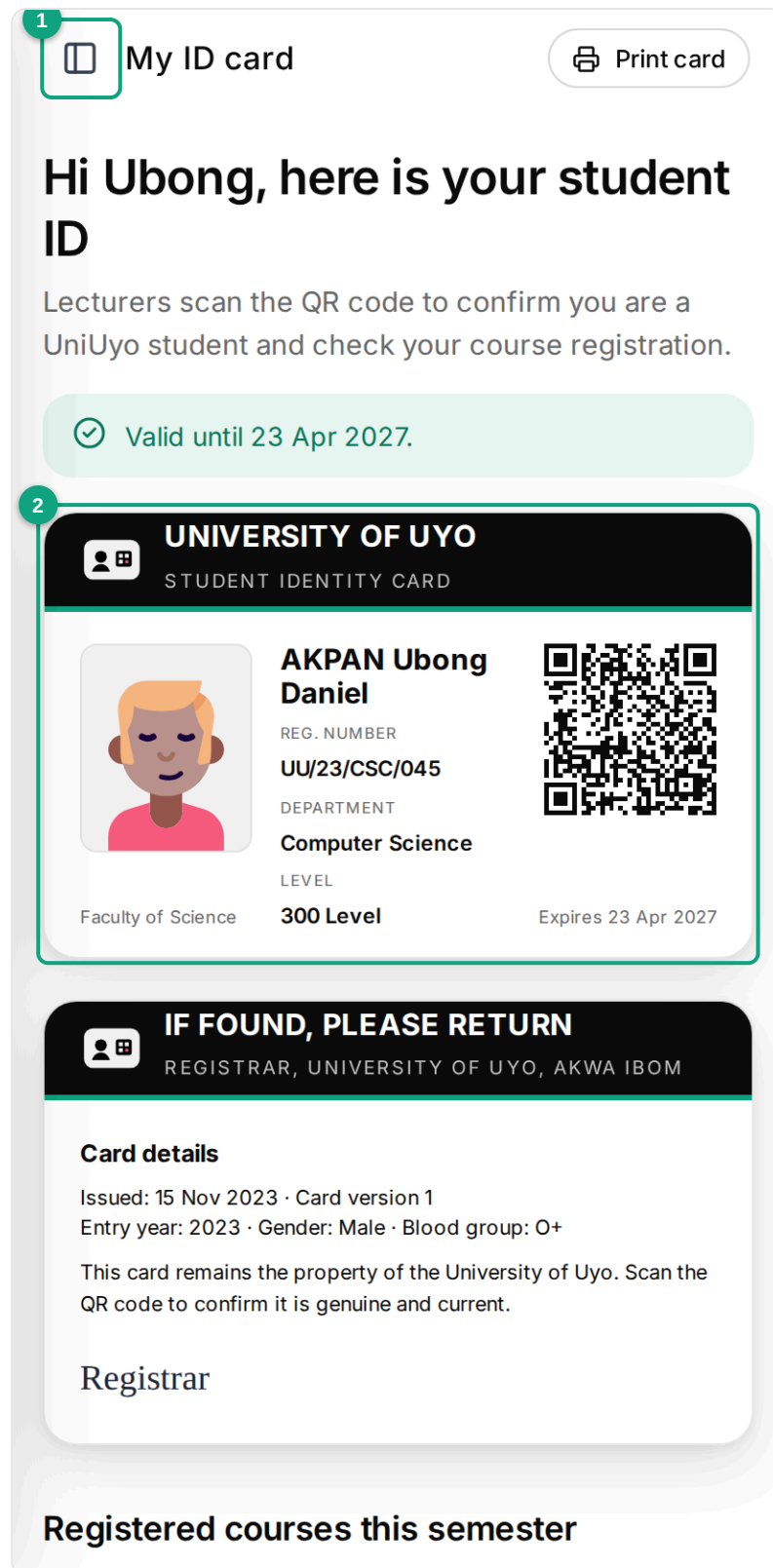


Opened from the QR code with any phone camera.

- 1 What any phone camera sees when it scans the card: a public page that only confirms the card is genuine and current.
- 2 Only details already printed on the card are shown. Class registration needs a lecturer login.

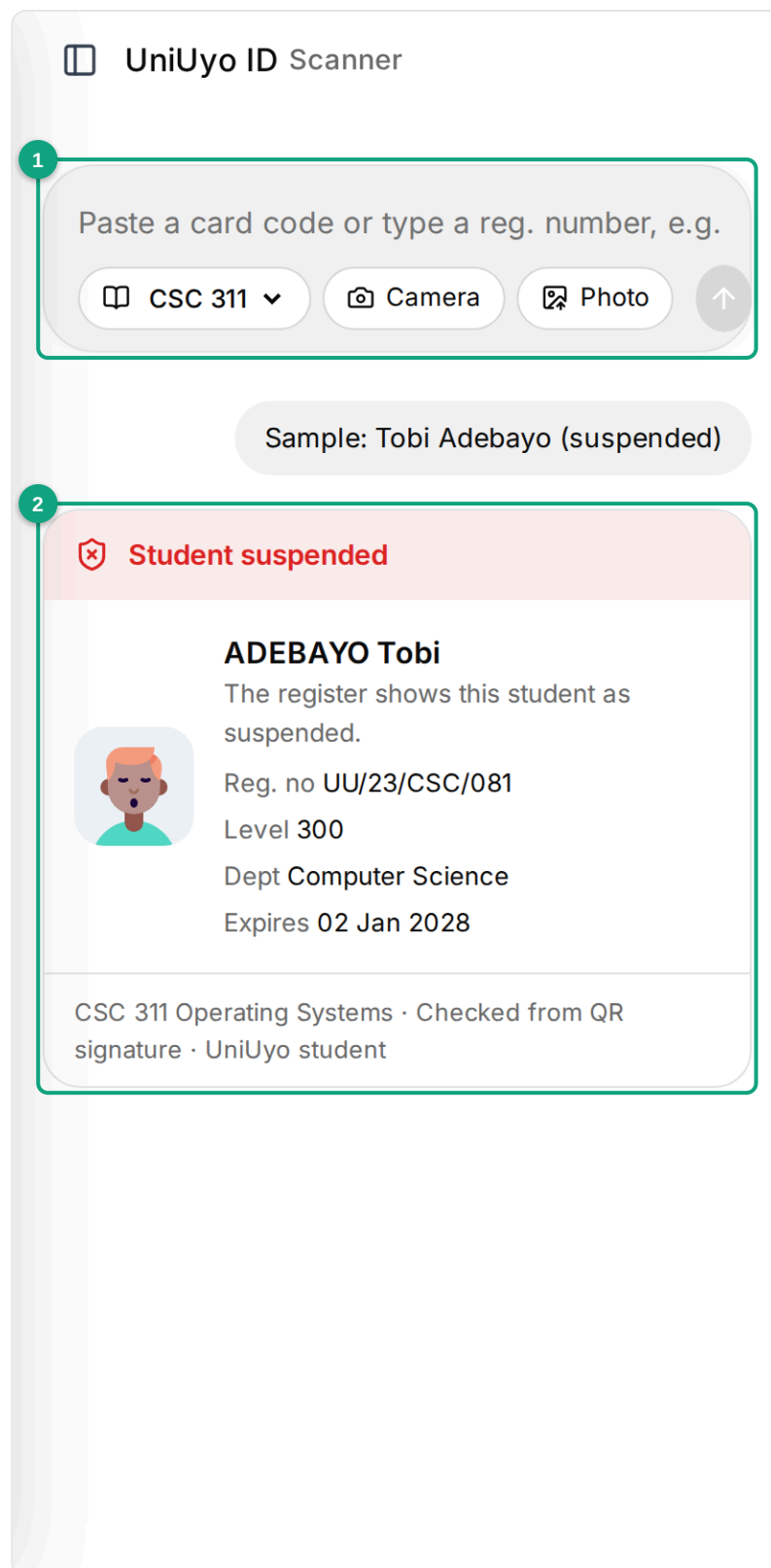
5. Mobile view

All grids use `minmax(0, 1fr)`. The e2e mobile test fails if the page scrolls sideways.



Student card on a phone.

- 1 The sidebar slides in from this button on phones.
- 2 The card scales with container query units so it stays the right shape at any width.



Lecturer scanning on a phone.

- 1 Composer with course, camera and photo buttons.
- 2 Red result: suspended student.

6. Running locally

```
service postgresql start
sudo -u postgres createdb uniid
sudo -u postgres createdb uniid_test
cp .env.example .env      # DATABASE_URL, SESSION_SECRET
npm install
npm run db:migrate
npm run db:seed
npm run dev                # http://localhost:3000
```

Script	What it does
dev / build / start	Development server, production build, production server
start:prod	Migrate, seed only when empty, next start -H 0.0.0.0 (Railway)
db:generate, db:migrate, db:seed	New migration, apply, reset and seed
test:unit, test:integration, test:e2e	Test suites
docs:pdf	Rebuild this PDF

The camera needs HTTPS or localhost. On the deployed site it works on phones and laptops.

7. Testing

19

unit tests passed

14

integration tests passed

7

e2e tests passed

Suite	Covers
Unit	Token format, round trip, edited reg number, bumped version, wrong key, junk input, URL extraction; every verdict and their order; card states; reg number normalising; dates; sessions.
Integration (Postgres)	Seed size; in class; full URL input; Law student; same department; suspended, expired, old and new card versions; forged and unknown; typed reg number; lecturer cannot scan for other courses; course and scan scoping; roster counts; student record scoping; public lookup; scan log.
End to end (Playwright)	Login prefill; student card, QR link and public page; lecturer scans a class member, a Law student and a forged code; QR photo upload decoded in the browser; inline error and class list; students blocked from lecturer pages; mobile card and mobile scan.

Latest run: unit 19, integration 14, 0 failed; e2e 7 of 7 passed (6 desktop, 1 mobile).

8. Deployment

Runs on Railway in the **school-projects** project as the **uni-digital-id** service with its own **uni-digital-id-postgres** database, deployed from the main branch of the GitHub repository.

Setting	Value
Build	npm run build
Start	npm run start:prod
Health check	/api/health (503 if Postgres is down)
DATABASE_URL	\${{uni-digital-id-postgres.DATABASE_URL}}
SESSION_SECRET	random value; also the key for QR signatures
NODE_ENV	production

9. Five minute presentation script

Time	What to do and say
0:00	Problem: lecturers cannot easily tell whether someone in class is a UniUyo student, or whether they belong to the course. Paper IDs are easy to fake. We use the existing student register and add a signed QR code.
0:30	On a phone, log in as the student (Ubong Akpan). Show the card: photo, name, reg number, department, expiry, QR, and the registered courses.
1:15	On the laptop, log in as the lecturer. Point out the sidebar: courses and recent scans. CSC 311 is selected.
1:40	Click Camera and scan the phone's card: green, Belongs to CSC 311, with the student's details.
2:20	Type the Law student's registration number UU/23/LAW/112: amber, UniUyo student, not in this class. This is the case the lecturer asked for.
2:50	Click the forged code sample: red, Not a UniUyo ID. Explain the HMAC signature in one sentence. Click the suspended sample: red.
3:30	Open CSC 311 in the sidebar: class list, verified today, flags for the suspended and expiring cards.
4:00	Scan the student's QR with the phone's normal camera app: the public page says Genuine UniUyo student ID.
4:20	Architecture and tests: Next.js Server Components and Actions, Drizzle with Postgres, pure verdict functions, test counts, Railway with health checks.
4:45	Next steps: attendance export per lecture, card reissue workflow at the ICT Centre. Questions.

Image credits

Campus and lecture photos are Creative Commons images found through Openverse, stored in `public/images` with credits in `public/images/credits.json`. Student photos are generated portraits (DiceBear), not real people.